CYBERJAYA 5G FUTURE TALENT INNOVATION CHALLENGE 2026

Track 2 · 5G Connected & Safe Last-Mile Mobility

JalanSelamat5G

Every rider a sensor. Every second counts.

A live 5G safety loop for Cyberjaya's motorcyclists. It warns drivers and riders of junction conflicts in milliseconds, calls 999 automatically after a crash, and shows the council which junction to fix first.

Team YO5G · Universiti Malaya

Organised by KOLAXUS

In collaboration with CYBERVIEW



Photo: Andrew Lawson, CC BY 2.0, via Wikimedia Commons

TEAM INFORMATION

Team YO5G, Universiti Malaya

Field	Details
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Track	Track 2: 5G Connected & Safe Last-Mile Mobility

Team members

1. **Tan You Tuan**
   Team leader · Mechanical Engineering
   5G networks, edge computing and sensing

2. **Low Li Xin**
   Electrical Engineering
   Phone sensors, signal processing, data pipeline

3. **Tan Hong Shen**
   Mechanical Engineering
   Crash dynamics: telling a real crash from a dropped phone

All members are registered students at Universiti Malaya

KOLAXUS Business Innovation Platform | CYBERVIEW

1. THE PROBLEM

Malaysia's riders are dying, and help arrives late

12

motorcyclists die on Malaysian roads every day

4,452 motorcyclist and pillion deaths in 2025 ÷ 365 days = 12.2 a day (other reports give 4,267–4,542)

+21% commuting accidents among workers, 2023 to 2025 (46,542 cases in 2025, PERKESO)



2 in 3

road deaths in 2025 were **motorcyclists or pillion riders** (4,452 of 6,581)

34%

of critical ambulance calls **reached the scene within 15 minutes** in 2025. Target: 50%.



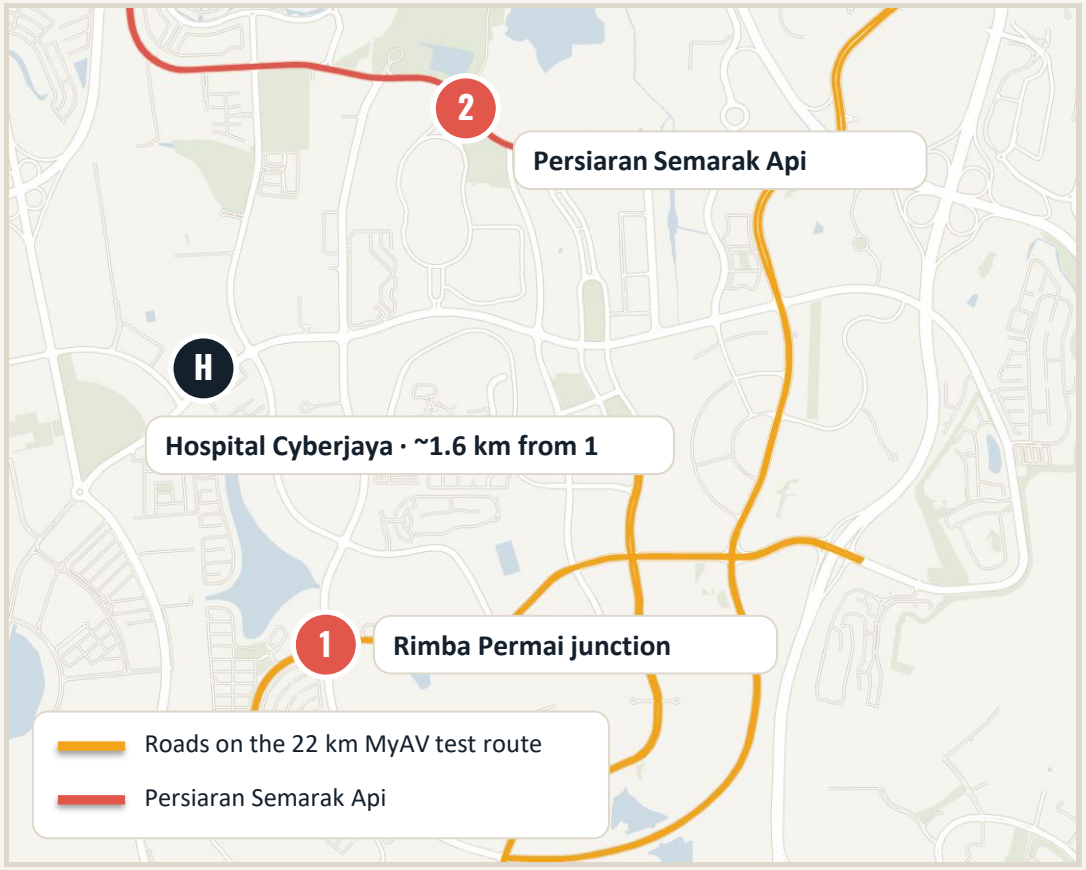
RM30 billion

lost to road crashes in 2025, MIROS estimate (lost output, treatment and family costs)

Sources: The Vibes, 21 Oct 2026 (PDRM 2025 report) · Malay Mail, 12 Oct 2026 · NST, Aug 2026 (MIROS) · Malay Mail, 23 May 2026 (PERKESO)

1. THE PROBLEM · IN CYBERJAYA

It is happening on Cyberjaya's own roads



Map data © OpenStreetMap contributors (ODbL)

1 The Rimba Permai "death junction"

Crashes almost every week; about 2,000 residents depend on it. MPSepang then appointed a contractor for smart traffic lights (26 Nov 2025). Utusan; Malaysia Tribune

2 6 Apr 2026 · Persiaran Semarak Api

A rider in his 20s was killed when a car turned right across his path. NST and Malay Mail, 7 Apr 2026

3 9 Sep 2026 · Cyberjaya–Putrajaya road

A rider in his 30s died after losing control at km 47.1. His pillion was injured. Bernama, 9 Sep 2026

173,129 crashes in Selangor in 2023, the most of any state (29% of Malaysia). **No crash count is published for Cyberjaya itself.**

Sources: Utusan, Nov 2025 · Malaysia Tribune, Dec 2025 · NST, Apr 2026 · Bernama, Sep 2026 · Malay Mail, 1 Jan 2024 (Bukit Aman) · NST, Oct 2026 (MyAV route)

1. THE PROBLEM · WHY IT PERSISTS

Today's tools react only after the damage is done

1 Too slow

A site becomes an official blackspot only after 3 similar crashes within 50 m in 3 years.

Evidence: hard braking shows up on 18× more road segments than reported crashes, so the warning signs exist much earlier (Google Research, Jan 2026).

2 One-way

A siren only warns drivers who hear it, and only in time if they react.

Evidence: in Dec 2025 two cars refused to give way to an ambulance on the Kesas Highway (Malay Mail).

3 Not built for small bikes

Crash calls exist for cars and premium motorcycles; C-V2X junction warnings need a radio in every car and bike.

Evidence: none is linked to MERS 999 for Malaysian riders, yet 2 in 3 road deaths are motorcyclists.

4 Unmeasured

RM118.5M treated 342 blackspots in 2021–2025, but we found no published before-and-after results.

Evidence: the Works Ministry gets about RM2B of the RM4B a year it needs, so it must pick hotspots carefully (Malay Mail, Aug 2026).

Our insight: riders already carry the sensors. Hard braking, swerves and falls can be measured within seconds, but nobody collects them in real time or acts on them.

Sources: The Star / Paultan, Jul 2026 (blackspot criteria) · Google Research, Jan 2026 · Malay Mail, Dec 2025 · Ducati (C-V2X demo) · Malay Mail, Aug 2026

2. TARGET USER

Meet Amir, 24, a food-delivery rider in Cyberjaya



Real rider in Penang, not Amir. Photo: S. H. Darlin, CC BY 2.0

HIS DAY

- Works the lunch and dinner peaks, often in rain
- Phone on the handlebar for orders, but unsafe to tap while riding
- Crosses Rimba Permai junction several times a day
- Fears crashing alone at night with no one to call

What he needs

1. Warnings he can hear before he reaches danger
2. Help that comes even when he cannot call
3. Proof that his near-misses lead to real fixes

Also benefits: 999 crews, councils, drivers

He is one of ~600,000 p-hailing riders. What published Malaysian studies show:

41%

of 225 delivery riders surveyed had crashed while delivering (Rusli et al., 2022)

88.8%

of p-hailing riders surveyed check or type on their phone while riding (Transport Ministry, 2023)

23.2%

average situational-awareness test score of 264 riders; the authors call for collision-warning tech (MIROS-led, 2026)

Amir is an illustrative persona · **Sources:** Rusli et al., 2022 · The Vibes, Apr 2023 · Ibrahim et al., 2026 · NST, Oct 2026 · Wikimedia Commons (photo)

3. OUR SOLUTION

JalanSelamat5G: one live safety loop, four steps

A safety kit inside the delivery apps riders already use, plus a safety engine on an edge server in Cyberjaya's 5G network.

STEP 1 **Sense**	Riders' phones (motion and GPS), dashcams, roadside AI cameras, ambulance GPS and rain sensors stream signals over 5G.	**For Amir:** His phone does the work. He never has to tap or report.
STEP 2 **Decide**	The edge server matches signals by place and time, then grades each event with clear rules (see slide 9).	**For Amir:** Fewer false alarms, so he trusts every warning.
STEP 3 **Act**	In milliseconds to seconds: conflict alerts to driver and rider, hazard warnings, an automatic MERS 999 alert, ambulance signal priority.	**For Amir:** He hears danger before he sees it. Help comes even if he cannot call.
STEP 4 **Prove**	Near-misses and crashes are mapped by place and hour. Fixes are ranked by risk removed per ringgit, then checked.	**For Amir:** The junction he fears gets fixed first.

Track 2 fit

- **SAFETY** Notifies riders of sudden route hazards in real time
- **MOVEMENT** Tracks riders and automates rapid response
- **AWARENESS** Gives traffic managers live risk data

Track 2 focus areas from the official participant briefing (23 Oct 2026) · Steps 1–3 take seconds; step 4 takes weeks

4. INNOVATION

Others solve one piece. We close the loop for riders

Existing approach	Detects crashes automatically	Warns nearby riders in seconds	Designed for motorcycles	Gives councils evidence to fix roads	In use in Malaysia
Waze (crowd reports)	—	Partly	—	Partly	**Yes**
MBSJ SENSA AI camera (2026)	—	Partly	—	Partly	**Yes**
REALRIDER app (UK)	**Yes**	—	**Yes**	—	—
Bosch Help Connect (Germany)	**Yes**	—	**Yes**	—	—
Audi–Ducati C-V2X demo (2018)	—	**Yes**	**Yes**	—	—
Google hard-braking data (cars)	—	—	—	**Yes**	—
PERKESO MYRIDERID (QR card)	—	—	**Yes**	—	**Yes**
JalanSelamat5G	**Yes**	**Yes**	**Yes**	**Yes**	Pilot proposed

What is genuinely new

No new hardware on vehicles

Riders' phones plus one camera per junction. C-V2X needs a radio in every car and bike.

Computed at the 5G edge

Junction conflict warnings are computed inside Cyberjaya, not a distant cloud.

One stream, three uses

Protect now, rescue in seconds, fix within weeks, with every fix measured.

5. HOW IT WORKS

How one signal becomes a warning, a rescue and a fix

INPUTS

- Rider phones: motion + GPS
- Dashcams: short clips
- Roadside AI cameras
- Ambulance GPS (MERS 999)
- Rain and flood sensors

5G NETWORK

Low latency
a round trip of milliseconds

High uplink
many video clips at once

Network slice
a reserved lane for emergency data

Massive IoT
thousands of sensors per km²

EDGE SERVER IN CYBERJAYA

Fuse
Match signals by place and time

Confirm
Apply the four rules below

Score
Grade severity and junction risk

OUTPUTS

Drivers and riders
Conflict alert, under 100 ms

MERS 999
Crash alert with location and a clip

Traffic signals
Green wave for the ambulance

MPSepang / JKR
Weekly ranked list of fixes

Conflict · camera only
Turning car + oncoming rider → both warned, target < 100 ms

Crash · phone + 10 s
Impact and fall, no reply to “Are you OK?” → MERS 999 alerted

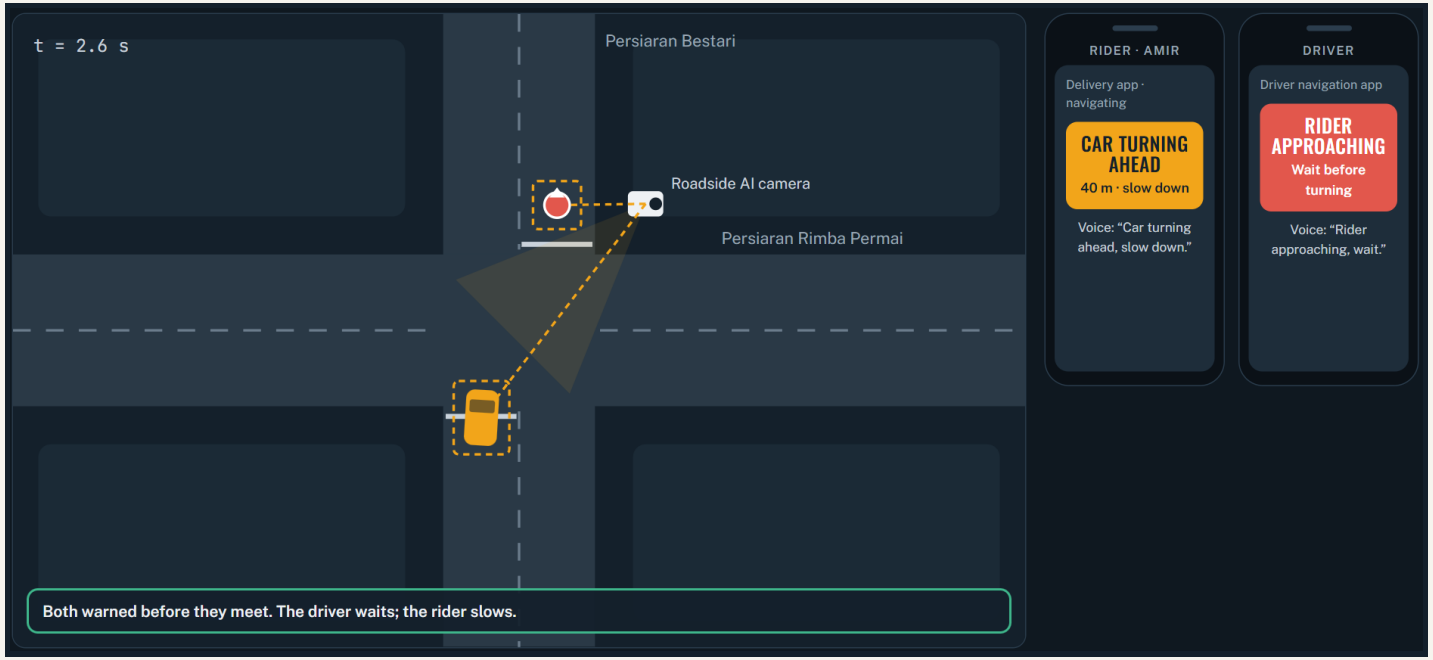
Hazard · 2+ sources
Confirmed within seconds → riders nearby warned in < 100 ms

Hotspot · weeks
Repeated near-misses → junction ranked for the council

Privacy by design: faces and plates are blurred on the phone; only short event clips leave the device

5. HOW IT WORKS · LIVE DEMO

See it work: our interactive prototype



Scenario 1 frozen at t = 2.6 s: the camera flags the turning car and both phones warn.

Four scenarios to try

1. **Junction conflict**
   Driver and rider both warned before they meet
2. **False alarm and hazard**
   Rider cancels a false alarm; two reports warn others
3. **Crash and 999**
   10-second check, then a simulated 999 alert
4. **Council view**
   Hotspot ranking and a before-and-after check

Open the demo

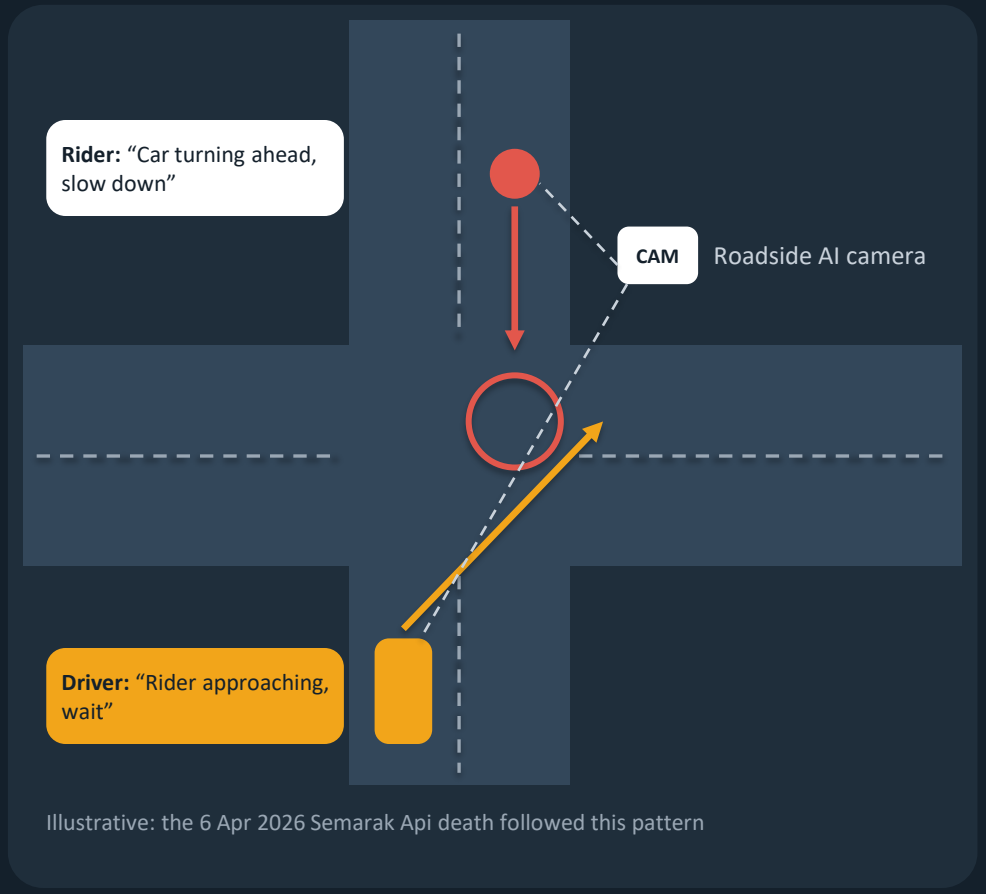
cynthialowlx.github.io/jalan-selamat-5g

Simulation: timings are design targets. It never contacts 999 or real traffic signals.

Built with plain HTML and JavaScript; map data © OpenStreetMap contributors · **Sources:** Source code and docs on GitHub

5. HOW IT WORKS · WHY 5G

5G warns the driver and the rider before they meet



Illustrative: the 6 Apr 2026 Semarak Api death followed this pattern

1. **Camera sees the conflict**
   A roadside AI camera at Rimba Permai spots a car starting a right turn as a rider approaches.
2. **Edge server predicts it**
   A server inside Cyberjaya's 5G network projects both paths. No second report is needed.
3. **Both get a warning**
   The driver's navigation app and the rider's phone sound an alert. Target: under 100 ms end to end.

Why milliseconds matter: at 60 km/h a rider covers 17 m every second

20 ms	**0.3 m**	3GPP limit for pre-crash sensing
100 ms	**1.7 m**	3GPP limit for V2X safety messages
1 second	**17 m**	A warning that arrives one second late

Latency is a design target to be measured in the pilot · **Sources:** 3GPP TS 22.185 (V2X latency requirements) · NST, Apr 2026

5. HOW IT WORKS · WHY 5G

What needs 5G, and what doesn't

Part of the loop	Could 4G do it?	Why 5G makes it work
Hazard map and weekly hotspot ranking	**Yes**	Not a 5G feature, and we say so. This part runs on any network.
Junction conflict warning in ~100 ms	**Not reliably**	Low latency plus an edge server in Cyberjaya keep camera-to-phone time short.
Video clips from many riders and cameras	**Strains capacity**	Higher uplink capacity per cell confirms events with images, not guesses.
Emergency data during a big incident	**Congests**	A network slice reserves a lane for emergency data (needs 5G Standalone; pilot with the telco).
Thousands of roadside sensors	**Limited**	5G is built for far more connected devices per square kilometre.

The network and the rules exist: 5G covers 82.4% of populated areas, and MCMC's 2025 C-V2X technical code already covers road-user warnings over the cellular network with edge computing.

Sources: Bernama, Aug 2025 (DNB coverage) · MCMC/MTSFB TC G049:2025

5. HOW IT WORKS · SCENARIO

6:40 pm, raining, at the Rimba Permai junction

10 min before

1

Three riders brake hard at the same spot. Riders approaching hear “Slippery junction ahead.”

0 s

2

A car turns across Amir’s path despite a conflict alert. His phone senses impact and a fall: “Are you OK?” No reply.

+10 s

3

MERS 999 gets his location and a short clip. Riders behind hear “Crash ahead, slow down.”

+1 min

4

The ambulance gets signal priority. Drivers ahead hear “Ambulance behind you, keep left.”

Weeks later

5

MPSepang gets a ranked fix for this junction, with near-miss counts before and after.

What changes, and for whom

Amir
Help is requested ~10 s after the crash, not when a passer-by happens to call

Riders behind
Warned before they reach the same spot

MERS 999 crews
Drivers keep left and lights turn green

MPSepang
A ranked, costed fix, then before-and-after proof

Illustrative scenario at a real junction · Today only 34% of critical ambulance calls are reached within 15 minutes · **Sources:** Malay Mail, 12 Oct 2026

6. IMPACT

Comparable systems show what each part can deliver

UP TO

40%

faster emergency response from automatic crash calls

European Commission eCall estimate (up to 50% in rural areas)

FIELD RESULT

14–23%

shorter emergency runs with traffic-signal priority

Denver field study (1978): about 70 s saved per run (US DOT ITS)

FIELD RESULT

60%

fewer critical conflicts after a data-guided signal change

Bellevue, USA: a US$10,000 fix checked with video analytics

EARLY WARNING

18×

more road segments show hard braking than have crash records

Google Research, Jan 2026: risk is visible before deaths

Our pilot targets

Rimba Permai junction and the MyAV route, 6 months. Targets to test, set below what comparable systems achieved.

< 60 s

from crash to MERS 999 alert (10 s cancel window + sending)

≥ 90%

of hazard warnings confirmed real (tiered rules)

-30%

near-misses after the first fix (half of Bellevue's 60%)

-15%

ambulance time through the junction (below Denver's 14–23%)

Sources: European Commission (eCall) · US DOT ITS (Denver) · Roadway Safety Foundation (Bellevue) · Google Research

7. FEASIBILITY

Most of the building blocks already exist in Cyberjaya

Already in place

- **Cyberview's 22 km MyAV test route** runs along Persiaran Rimba Permai; a driverless van trial began there in Oct 2026.
- **Malaysia's C-V2X technical code** (MCMC/MTSFB, Feb 2025) covers road-user warnings via the cellular network and edge computing; Cyberview sat on its working group.
- **MBSJ's SENSA AI camera** already gives real-time audio and visual warnings at a Subang Jaya crossing (Mar 2026).
- **MPSepang appointed a contractor** for smart lights at Rimba Permai (Nov 2025). Our data can show whether they cut near-misses.
- **Malaysia's first remote operations centre** for autonomous vehicles opened in Cyberjaya (Jul 2026).
- **MOH is strengthening MERS 999** and buying 400 ambulances under Budget 2027.

Proposed partners and their role

Partner	Role in the pilot
Cyberview	Living lab, MyAV route, data-sharing agreement
MPSepang and JKR	Own the roads; act on ranked fixes
CelcomDigi / U Mobile	5G network, edge server, network slice
Grab, Foodpanda, Lalamove	Add our safety kit to rider apps
MOH (MERS 999), PDRM, Bomba	Receive automatic crash alerts
MIROS and Universiti Malaya	Validate the data and the evaluation

Sources: NST, Oct 2026 · MCMC/MTSFB TC G049:2025 · SAYS, Mar 2026 · Malaysia Tribune, Dec 2025 · Malay Mail, Jul 2026 · Bernama, Oct 2026

7. FEASIBILITY · WHO PAYS AND THE RISKS

Who pays, and how we handle the risks

Councils and JKR subscribe

The Works Ministry gets about RM2B of the RM4B it needs each year. Our ranking makes each ringgit count.

Delivery platforms join

Under the Gig Workers Act, platforms must register every rider for SOCSO cover. Safer riders mean fewer lost delivery hours.

Telcos gain a showcase

A flagship public-safety use case for edge computing and network slicing.

Riders use it free.

Assumptions we test first: platforms embed the kit · MERS 999 accepts machine alerts · MPSepang controls the junction signals

Risk	How we handle it
Privacy: PDPA fines now reach RM1M	Processed on the phone; faces and plates blurred; only short event clips uploaded, deleted after 30 days.
False alarms: one US county logged 155 false phone crash calls	Tiered rules, a 10-second “Are you OK?” cancel, and a confidence score for dispatchers.
Rider distraction	Voice and vibration only. Nothing to read or tap.
Low adoption	Built into apps riders already use; they opt in once.
Platforms decline to embed it	Fallback: a standalone app promoted through PERKESO's MyRider campaign.
5G Standalone not yet everywhere	Pilot on prioritised 5G; add network slicing with the telco in phase 3.

Sources: Malay Mail, Aug 2026 · ISEAS, Jul 2026 (Gig Workers Act) · DFDL (PDPA) · Fox Business (false calls) · Bernama (MyRider)

7. IMPLEMENTATION

Start at one junction, prove it, then scale

Months 0–3

1. Prepare

- Sign partner agreements
- PDPA privacy assessment
- Film a junction baseline

We measure: Near-misses per hour, before any change

Months 3–9

2. Pilot

- Rimba Permai + MyAV route
- 300 volunteer riders
- 4 roadside AI cameras

We measure: Share of alerts that are real; seconds from crash to 999

Months 9–18

3. Cyberjaya-wide

- All delivery platforms
- Live link to MERS 999
- Ambulance green wave

We measure: Ambulance travel time; crashes per 100,000 km

Month 18+

4. Scale

- Other Selangor towns
- National traffic centre
- Cell Broadcast alerts

We measure: Hotspots fixed per RM1 million spent

Gate 1: partners and privacy signed off

Gate 2: alerts proven real

Gate 3: crashes measurably fall

Pilot resources: riders' own phones · 4 roadside cameras at one junction · 1 edge-server slot from the telco · 1 data agreement with Cyberview

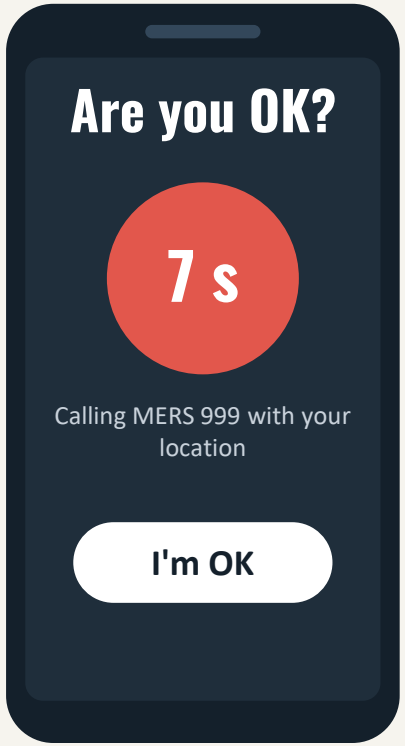
Each gate must be passed before the next phase starts

7. VISUAL CONCEPT

What riders, drivers and the council will see



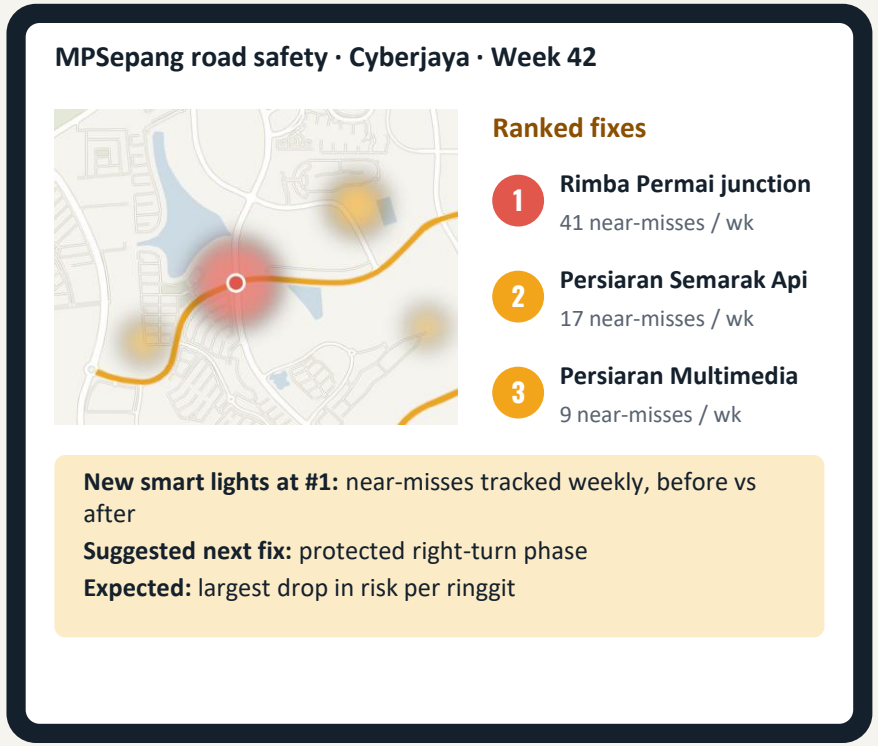
Rider: hears, never reads



Crash: help without a call



Drivers: clear the way



Council: fix what the data ranks first

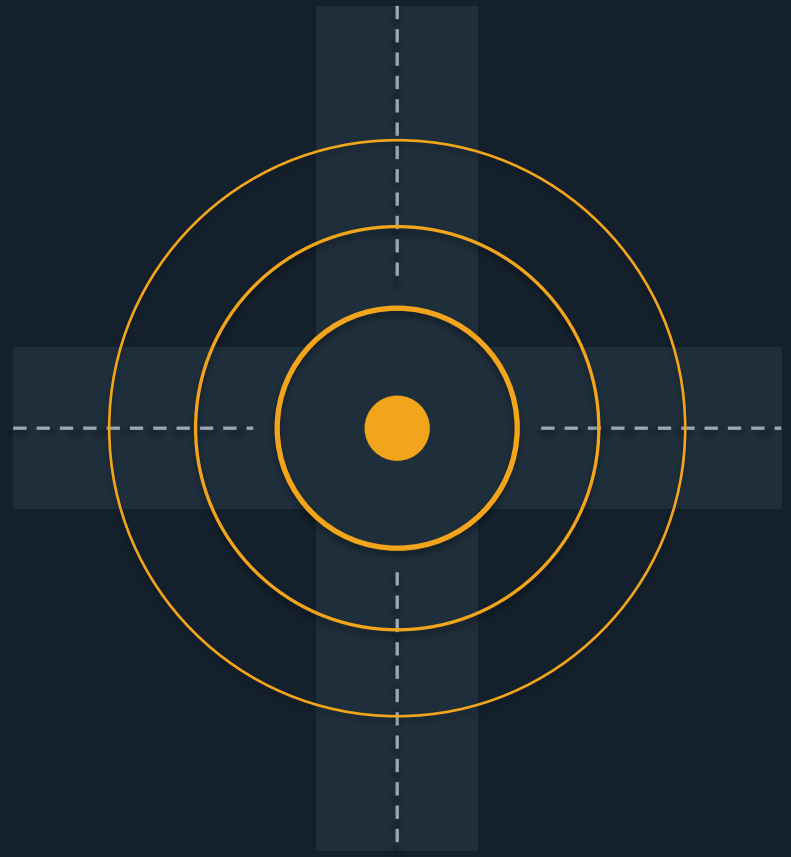
Concept mockups with illustrative data. Not screenshots of a working product.

Dashboard map drawn from OpenStreetMap data (ODbL)

8. WHY YOUR TEAM SHOULD WIN

Every rider a sensor. Every second counts.

Twelve motorcyclists die on Malaysia's roads every day, and Cyberjaya's own junctions are part of that toll. A 2026 MIROS-led study calls for collision-warning technology for riders. JalanSelamat5G delivers it with no new hardware on any vehicle: roadside cameras and riders' phones, linked by 5G and an edge server in Cyberjaya, warn drivers and riders of junction conflicts within a tenth of a second (our target), alert 999 after a crash, and show councils which fix saves the most lives. It fits Malaysia's 2025 C-V2X technical code. We start at one junction and let the data prove every claim.



99 words

Our ask: pilot partners: Cyberview, MPSepang, one telco and one delivery platform

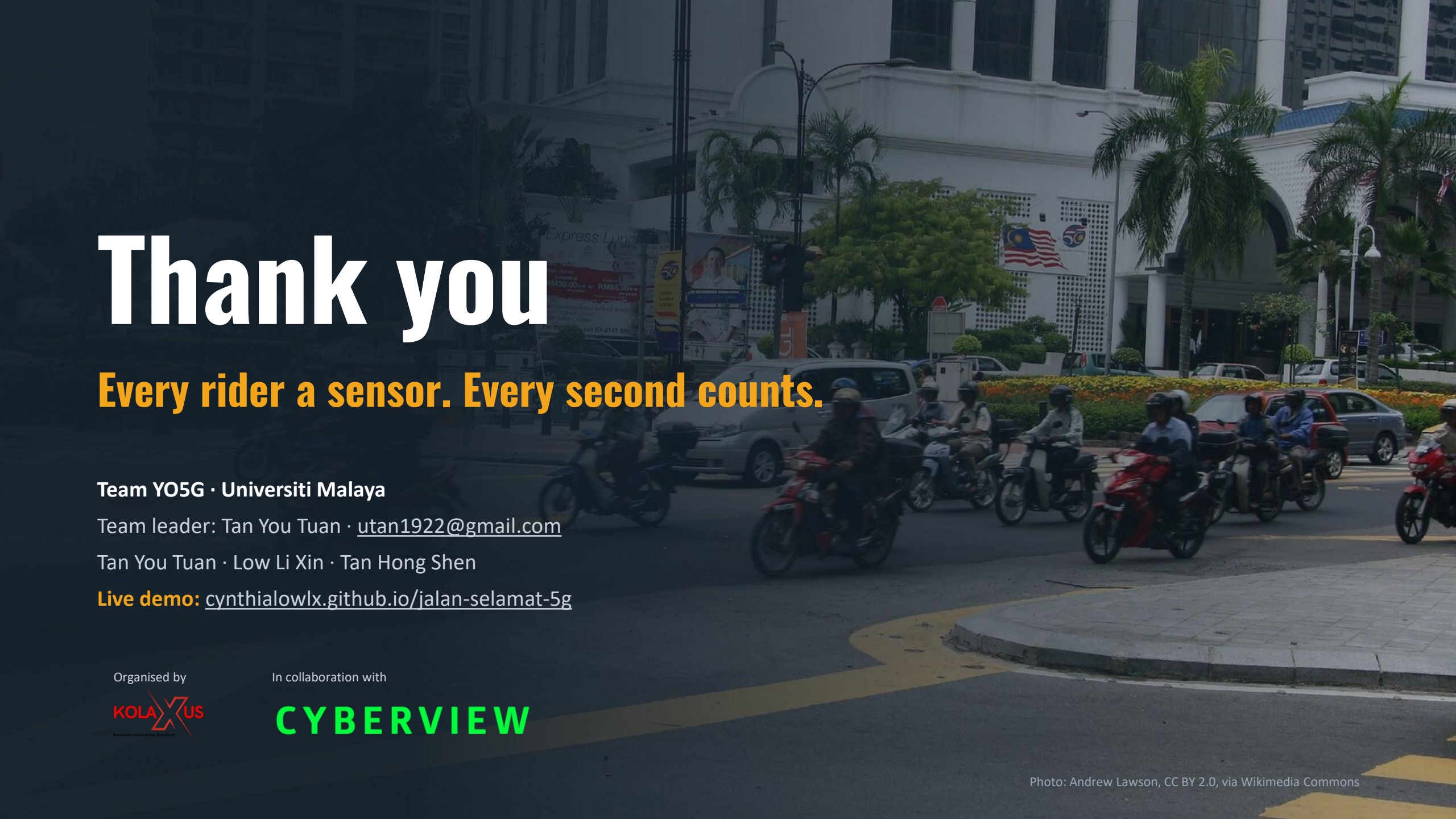
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Sources, image credits and AI use

1. **The Vibes, 21 Oct 2026.** Motorcyclists account for two-thirds of road deaths, 4,452 fatalities in 2025
2. **Malay Mail, 12 Oct 2026.** Only 34pc of emergency ambulance calls reach scene in under 15 minutes
3. **Bernama, 12 Oct 2026.** Ambulance response, MERS 999 and 400 new ambulances (Budget 2027)
4. **NST, Aug 2026.** Road crashes costing Malaysia RM30b yearly (MIROS)
5. **Malay Mail, 23 May 2026.** SOCSO: 46,542 commuting accidents last year
6. **Utusan Malaysia, 27 Nov 2025.** Simpang Rimba Permai, Cyberjaya jadi “simpang maut”
7. **NST, 7 Apr 2026.** Motorcyclist killed in Cyberjaya crash with car
8. **Bernama, 9 Sep 2026.** Rider killed on the Cyberjaya–Putrajaya road (km 47.1)
9. **Malaysia Tribune, 1 Dec 2025.** Rimba Permai: smart traffic lights within two months, says MPSepang
10. **Malay Mail, 1 Jan 2024.** Bukit Aman: 598,635 road accidents recorded in 2023
11. **Malay Mail, 10 Dec 2025.** Police hunt drivers who blocked ambulance on Kesas Highway
12. **Google Research, 13 Jan 2026.** Hard-braking events as indicators of road segment crash risk
13. **Paultan, 23 Jul 2026.** Works ministry identifies 94 locations for road safety upgrades
14. **Malay Mail, 18 Aug 2026.** Works ministry spends entire RM2b road maintenance allocation
15. **Rusli et al., 2022.** Case Studies on Transport Policy: food delivery riders with and without crash experience in Malaysia
16. **The Vibes, 5 Apr 2023.** Over 88% of p-hailing riders use their phones while riding: Loke
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22. **Bernama, 16 Aug 2026.** PERKESO MYRIDERID QR emergency identity for riders
23. **Ducati, 2018.** Audi and Ducati C-V2X demo: car turning across a motorcycle
24. **MCMC / MTSFB, Feb 2025.** TC G049:2025 C-V2X system architecture and specifications
25. **SAYS, 12 Mar 2026.** MBSJ launches SENSA AI pedestrian-warning camera in Subang Jaya
26. **3GPP TS 22.185.** Service requirements for V2X services (latency limits)
27. **Bernama, Aug 2025.** 5G coverage at 82.4% of populated areas (DNB)
28. **NST, 20 Oct 2026.** Driverless RoboVan begins six-month trial in Cyberjaya (MyAV route)
29. **Malay Mail, 31 Jul 2026.** Cyberjaya gets Malaysia's first autonomous vehicle control centre
30. **NST, Oct 2026.** Cell Broadcast System in final phase, live testing on Nov 19
31. **European Commission, 2013.** eCall: up to 40% faster response in cities, 50% in rural areas
32. **US DOT ITS Knowledge Resources.** Denver signal preemption: 14–23% shorter response
33. **Roadway Safety Foundation.** Bellevue, WA: Video Analytics Towards Vision Zero
34. **Fox Business.** Apple crash detection triggers false 911 calls (155 in one county)
35. **Kolaxus and Cyberview, 2026.** Challenge microsite and participant briefing (Track 2 focus areas)
36. **DFDL.** Personal Data Protection (Amendment) Act 2024: penalties up to RM1M

Images: cover photo Andrew Lawson (CC BY 2.0); rider photo Sharon Hahn Darlin (CC BY 2.0), both via Wikimedia Commons; maps drawn by the team from © OpenStreetMap contributors data (ODbL). All diagrams and mockups are original.

AI use: AI tools helped us research, edit and lay out this deck. The idea is the team’s own, and every figure was checked against the sources above.



Thank you

Every rider a sensor. Every second counts.

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Tan You Tuan · Low Li Xin · Tan Hong Shen

Live demo: cynthialowlx.github.io/jalan-selamat-5g

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